Consultancy Report: Designing a Retail Analytics Solution for Olist

To: Executive Leadership Team, Olist

From: Analytics Consultant Jorge Luis Del Angel Maldonado

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Subject: End-to-End Analytics Modernization using Google BigQuery

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1. Business Overview & Analytical Objectives

1.1 Organisation Profile

Olist is a prominent Brazilian e-commerce marketplace integrator that enables small and medium-sized businesses to sell their products through the country's largest retailers. Functioning as a "store of stores," Olist manages thousands of independent sellers, handling the entire transaction lifecycle—from product listing and payment processing to logistics coordination and customer service.

1.2 Key Operational Challenges

The organisation currently faces several friction points that limit its scalability:

- **Logistics Complexity:** Managing a vast network of sellers and delivery routes across Brazil leads to significant variability in delivery times.
- **Seller Performance Tracking:** With a fragmented seller base, identifying underperforming vendors before they impact customer satisfaction is a manual and reactive process.
- **Data Silos:** The existing OLTP system is optimized for transactions, making it difficult to perform complex, cross-functional analysis without impacting operational performance.
- **Payment Resolution:** Navigating Brazilian payment preferences, such as high-installment credit card purchases, requires robust financial tracking.

1.3 Analytical Questions

The proposed solution aims to answer the following strategic questions:

1. **Revenue Growth:** What is the monthly Gross Merchandise Value (GMV) and its growth rate across different product categories?
2. **Logistics Efficiency:** How does the "delay time" (actual vs. estimated delivery) vary by state, and what is its impact on customer satisfaction?

3. **Seller Success:** Which sellers are maintaining the highest order volumes and shortest lead times?
4. **Financial Health:** What is the distribution of payment types (e.g., credit card, boleto) and their impact on transaction values?

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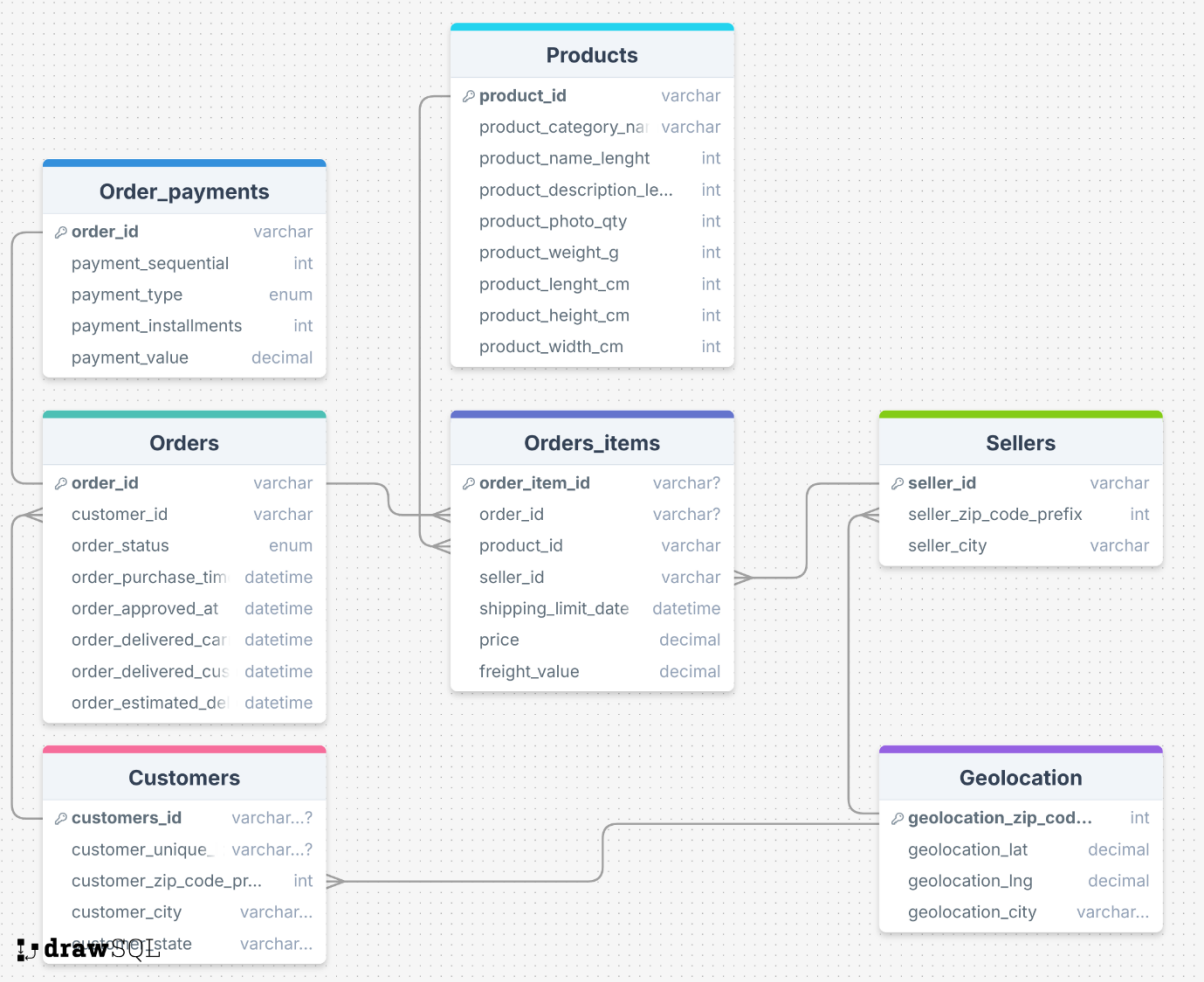
2. Data Sources & Ingestion Strategy

2.1 Internal Data Sources

The primary data source is the Olist OLTP MySQL database. The relevant tables include:



- **Customers & Sellers:** Demographic and geographic data.
- **Orders & Order_items:** Core transactional data including timestamps, pricing, and freight values.
- **Order_payments:** Detailed financial records of installments and payment methods.
- **Products:** Specifications such as weight, dimensions, and categories.

- **Geolocation:** Spatial data mapping zip codes to latitude and longitude.

2.2 Ingestion Approach

To modernize the pipeline, Olist will adopt an **ELT (Extract, Load, Transform)** strategy, which is well-suited for cloud-native analytics:

- **Ingestion Method:** Using **Google Cloud Data Fusion** or **BigQuery Data Transfer Service**, raw data will be extracted from the MySQL source and loaded into a "Staging" dataset within BigQuery.
- **Format:** Data will be landed in BigQuery tables as raw replicas of the source schema to maintain a "Bronze" layer of data.
- **Frequency:** Sales and order data will be ingested daily, while dimension tables like Products and Geolocation will undergo weekly full refreshes to capture updates.

2.3 Strategic Rationale

- **Separation of Concerns:**
  By landing raw data in a staging area, Olist can decouple extraction from transformation, reducing risk and improving data quality.
- **Scalability:**
  The use of serverless tools (BigQuery, Data Fusion) allows Olist to scale ingestion as data volumes grow, without manual intervention.
- **Data Freshness:**
  Daily ingestion of transactional data ensures that analytics and dashboards reflect the latest business activity.
- **Schema Evolution:**
  Weekly refreshes for dimension tables accommodate changes in product attributes and geographic mappings, supporting accurate reporting.
- **Compliance & Traceability:**
  Maintaining raw replicas and a clear ingestion schedule supports auditability and compliance with data governance standards.

2.4 Implications for Analytics

- **Enables Advanced Analytics:**
  The ELT approach and robust ingestion pipeline lay the groundwork for complex, cross-functional analysis—such as revenue growth tracking, logistics optimization, and seller performance monitoring.

- **Supports Data Quality:**
  By ingesting data in its raw form and transforming it in BigQuery, Olist can implement rigorous data quality checks and feature engineering before data reaches the analytics layer.

- **Facilitates Future Expansion:**
  The modular ingestion strategy makes it easier to onboard new data sources or adjust refresh frequencies as business needs evolve.

The "Data Sources & Ingestion Strategy" section is a critical pillar of Olist's analytics modernization. It ensures that all relevant business data is captured, ingested efficiently, and made available for high-performance analytics in BigQuery. This approach supports scalability, data quality, and compliance, positioning Olist to unlock deeper insights and drive business growth.

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3. Data Modelling

Data modelling is the backbone of Olist's analytics modernization. It transforms operational data into a format optimized for high-performance analytics, enabling the business to extract actionable insights at scale. The transition from a normalized OLTP (Online Transaction Processing) structure to a denormalized star schema is central to this transformation, supporting both speed and flexibility in reporting and analysis.

3.1 Conceptual and Logical Models

Conceptual Model

At its core, the conceptual model organizes Olist's business data around three major domains:

- **Sales Events:** Every transaction, from order placement to payment and delivery.
- **Logistics:** The entire shipping lifecycle, including delivery times and status.
- **Entities:** The key actors—customers, sellers, and products—that interact within the marketplace.

This model clarifies how different business processes interrelate, laying the groundwork for robust analytics.

Logical Model

The logical model translates these concepts into specific database relationships:

- **Primary Keys:** Unique identifiers for each entity (e.g., customer_id, seller_id, product_id).

- **Foreign Keys:** Linkages that enable joining data across tables (e.g., order_id in sales fact table referencing customer_id in the customer dimension).

This structure ensures that every analytical query can accurately connect sales, logistics, and entity data for comprehensive analysis.

3.2 Star Schema Design

The star schema is a proven data warehousing architecture that balances simplicity, performance, and analytical power. Olist's implementation features:

Fact Tables

- **Fact_Sales:**
    - Stores granular transactional metrics: price, freight value, payment value.
    - Captures the relationships between products, sellers, and customers for every item sold.
    - Enables analysis of sales performance, revenue growth, and seller success.
- **Fact_Delivery:**
    - Focuses on the shipping lifecycle: lead time (days), delay time, delivery status.
    - Supports logistics optimization and customer satisfaction analysis.

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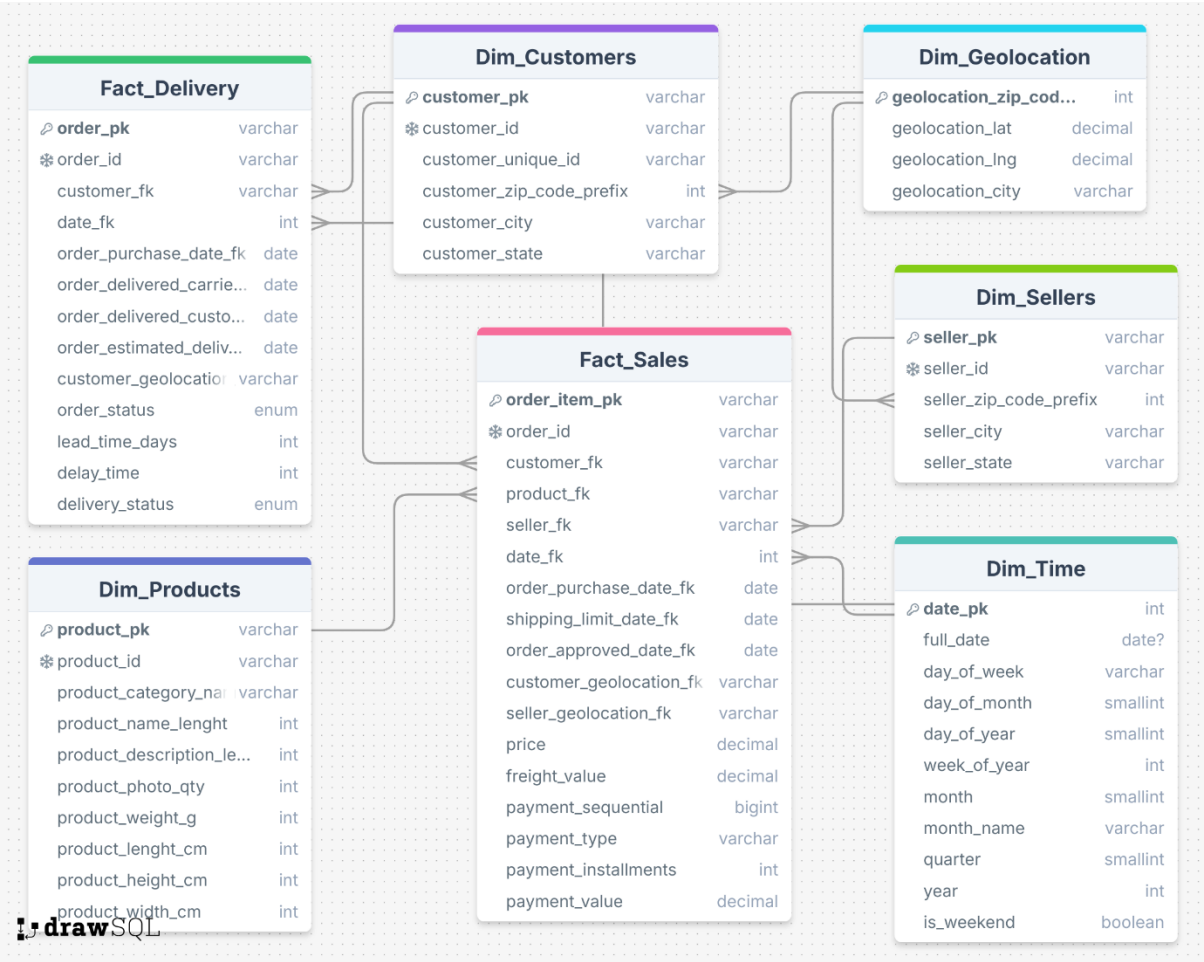

Dimension Tables

- **Dim_Customers & Dim_Sellers:**
  - Store identity and geographic details for buyers and vendors.
  - Allow segmentation and performance tracking by region or seller type.
- **Dim_Products:**
  - Contains product attributes (weight, dimensions) and category hierarchies.
  - Supports product-level analytics and inventory management.
- **Dim_Geolocation:**
  - Aggregates spatial data by zip code prefix, providing average coordinates for mapping.
  - Enables geographic analysis of sales and delivery performance.
- **Dim_Time:**
  - Centralizes time-series data, allowing aggregation by week, month, quarter, and year.
  - Facilitates trend analysis and seasonality studies.

3.3 Data Warehouse Layer ("Gold Layer")

The star schema forms the “Gold Layer” of Olist’s data warehouse in BigQuery. This layer is designed for:

- **High-Performance Analytics:**
  Denormalized tables minimize the need for complex joins, speeding up query execution.
- **Flexible Reporting:**
  Dimension tables enable slicing and dicing of data across multiple business perspectives.
- **Scalability:**
  The schema supports growing data volumes and evolving business requirements

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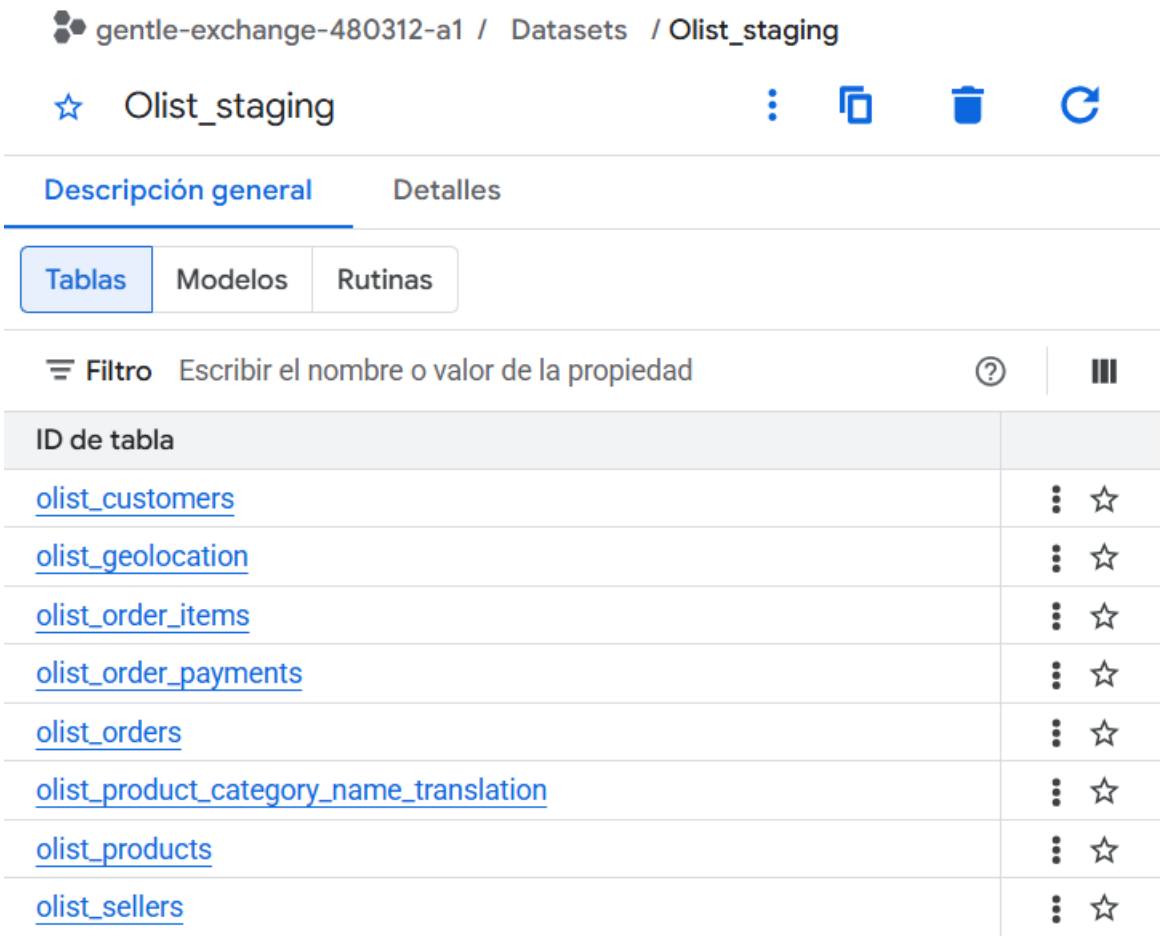
4. ETL/ELT Pipeline Design

The pipeline follows a structured sequence to ensure data quality and analytical readiness.

4.1 Transformation & Loading Sequence

1. **Staging:** Data is loaded from MySQL into olist_staging tables in BigQuery.



2. **Transformation and Feature Engineering**

Before data reaches the warehouse layer, critical transformations are performed:

- **Feature Engineering:**
  - Metrics like lead_time_days are calculated in the staging layer (e.g., difference between delivery date and purchase timestamp).
  - Surrogate keys (UUIDs) are generated to ensure uniqueness and simplify joins.
- **Data Quality Checks:**
  - Automated rules ensure non-negative values for price and freight.
  - Referential integrity is enforced: every order_id in the fact table must have corresponding entries in the dimension tables.

Feature engineeri... Ejecutar

```sql
ALTER TABLE `gentle-exchange-480312-a1.Olist_staging.olist_orders`
ADD COLUMN lead_time_days INT64,
ADD COLUMN delay_time INT64,
ADD COLUMN delivery_status STRING;

UPDATE `gentle-exchange-480312-a1.Olist_staging.olist_orders`
SET
  lead_time_days = DATETIME_DIFF(
    order_delivered_customer_date, order_purchase_timestamp, DAY)
WHERE TRUE;

UPDATE `gentle-exchange-480312-a1.Olist_staging.olist_orders`
SET
  delay_time = DATETIME_DIFF(
    order_delivered_customer_date, order_estimated_delivery_date, DAY)
WHERE TRUE;

UPDATE `gentle-exchange-480312-a1.Olist_staging.olist_orders`
SET
  delivery_status = CASE
    WHEN order_delivered_customer_date IS NULL THEN 'Pending'
    ELSE 'Delivered'
    END
WHERE TRUE;
```

3. **Data Migration:** Data is then moved into the Olist_dw (Data Warehouse) schema, where UUIDs are generated for surrogate keys.

Star_schema_data_migration Ejecutar

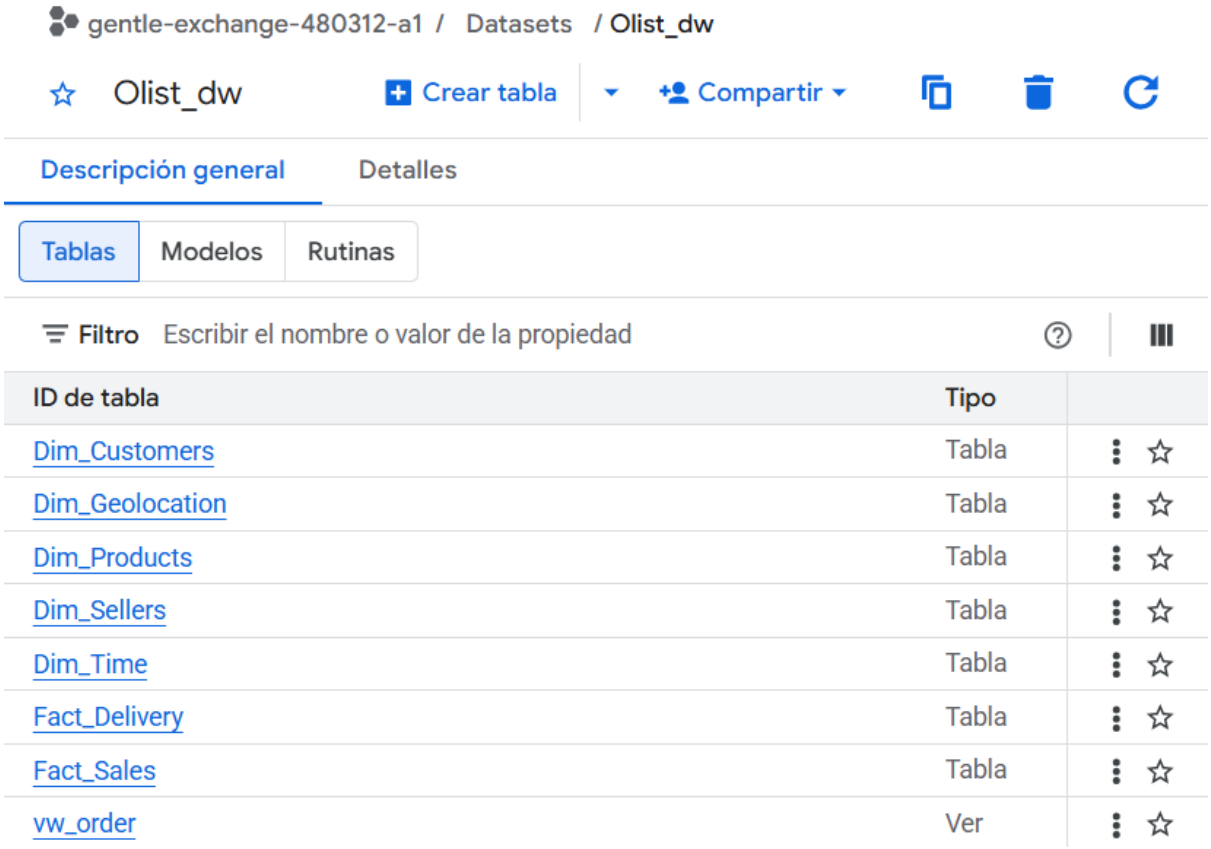
```sql
-- Create Dimension Tables

-- Dim_Customers
CREATE OR REPLACE TABLE `gentle-exchange-480312-a1.Olist_dw.Dim_Customers`(
  customer_pk STRING NOT NULL
  OPTIONS (description = "Unique identifier for the customer dimension"),
  customer_id STRING NOT NULL
  OPTIONS (description = "Original customer ID from staging"),
  customer_unique_id STRING
  OPTIONS (
    description = "Unique identifier for a customer across multiple purchases"),
  customer_zip_code_prefix STRING
  OPTIONS (description = "First 5 digits of customer's zip code"),
  customer_city STRING
  OPTIONS (description = "Customer's city"), customer_state STRING
  OPTIONS (description = "Customer's state"));

-- Dim_Products
CREATE TABLE IF NOT EXISTS `gentle-exchange-480312-a1.Olist_dw.Dim_Products`(
  product_pk STRING NOT NULL
```

4.2 Indicative SQL for Key Steps

Populating Fact_Sales (Warehouse Layer):



Olist's data modelling strategy—anchored in a star schema—transforms fragmented operational data into a unified, analytics-ready asset. This approach empowers Olist to analyze sales, logistics, and seller performance with speed, accuracy, and flexibility, driving better business outcomes in the competitive Brazilian e-commerce market

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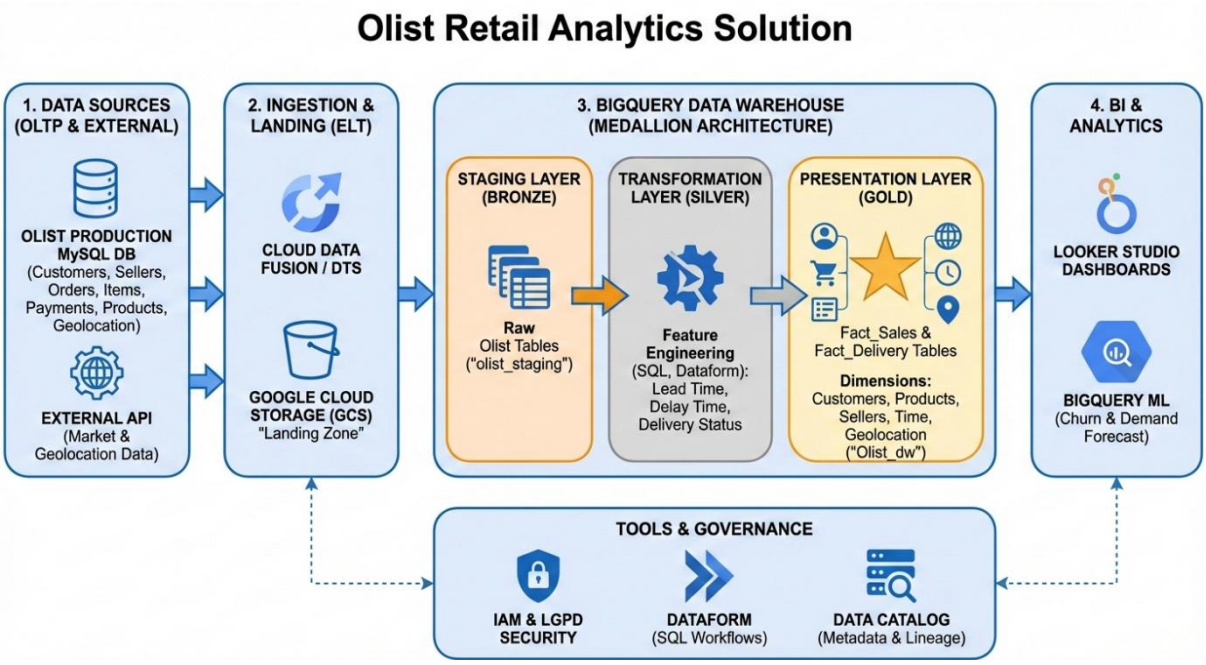
5. Cloud Architecture Using GCP

The architecture leverages serverless components to ensure scalability and cost-efficiency.

Olist's cloud architecture is designed to deliver scalable, cost-efficient, and secure analytics by leveraging the full suite of Google Cloud Platform (GCP) services. The architecture supports the entire analytics lifecycle—from raw data ingestion to advanced business intelligence—while ensuring operational flexibility and robust governance.

5.1 Architecture Components



Core Architecture Components

a. Data Sources (OLTP & External)

- **Olist Production MySQL Database:**
  The primary source of operational data, including customers, sellers, orders, payments, and products.
- **External API (Geolocation Data):**
  Supplements internal data with spatial information for advanced geographic analysis.

b. Ingestion & Landing (ELT)

- **Google Cloud Storage (GCS):**
  Serves as the landing zone for raw data files before ingestion. This decouples extraction from transformation, supporting traceability and compliance.
- **Google Cloud Data Fusion / BigQuery Data Transfer Service:**
  Automates the extraction and loading of data from MySQL and other sources into BigQuery staging tables, forming the “Bronze” layer of the data warehouse.

c. BigQuery Data Warehouse (Medallion Architecture)

- **Staging Layer (Bronze):**
  Raw replicas of source tables, preserving schema fidelity and supporting auditability.

- **Transformation Layer (Silver):**
  Feature engineering, data quality checks, and surrogate key generation are performed here.
- **Presentation Layer (Gold):**
  Denormalized fact and dimension tables (star schema) optimized for high-performance analytics and flexible reporting.

d. Dataform

Manages SQL workflows, orchestrates transformations, and maintains documentation within BigQuery. This ensures repeatability and transparency in data engineering processes.

e. Looker Studio

The business intelligence (BI) tool for dashboarding and data exploration. It connects directly to BigQuery's presentation layer, enabling real-time analytics and interactive visualizations for business users.

5.2 Management & Cost Considerations

- **Storage & Compute:** BigQuery's separation of storage and compute allows Olist to pay only for the storage used and the data scanned during queries.
- **Access Control:** Managed via **Identity and Access Management (IAM)**, Granular permissions ensure that data engineers have full access to all layers, while business analysts are restricted to the presentation layer. This protects sensitive data and enforces governance policies.
- **Cost Management:** We will implement table partitioning on order_purchase_date_fk to reduce the amount of data scanned, thereby lowering query costs. Partitioning tables by order purchase date reduces the amount of data scanned during queries, significantly lowering costs and improving query performance.

Olist's GCP-based cloud architecture is a robust, future-proof foundation for retail analytics. By integrating serverless data warehousing, automated workflows, and advanced BI tools, Olist achieves scalable, secure, and cost-effective analytics—empowering business leaders to make data-driven decisions with confidence.

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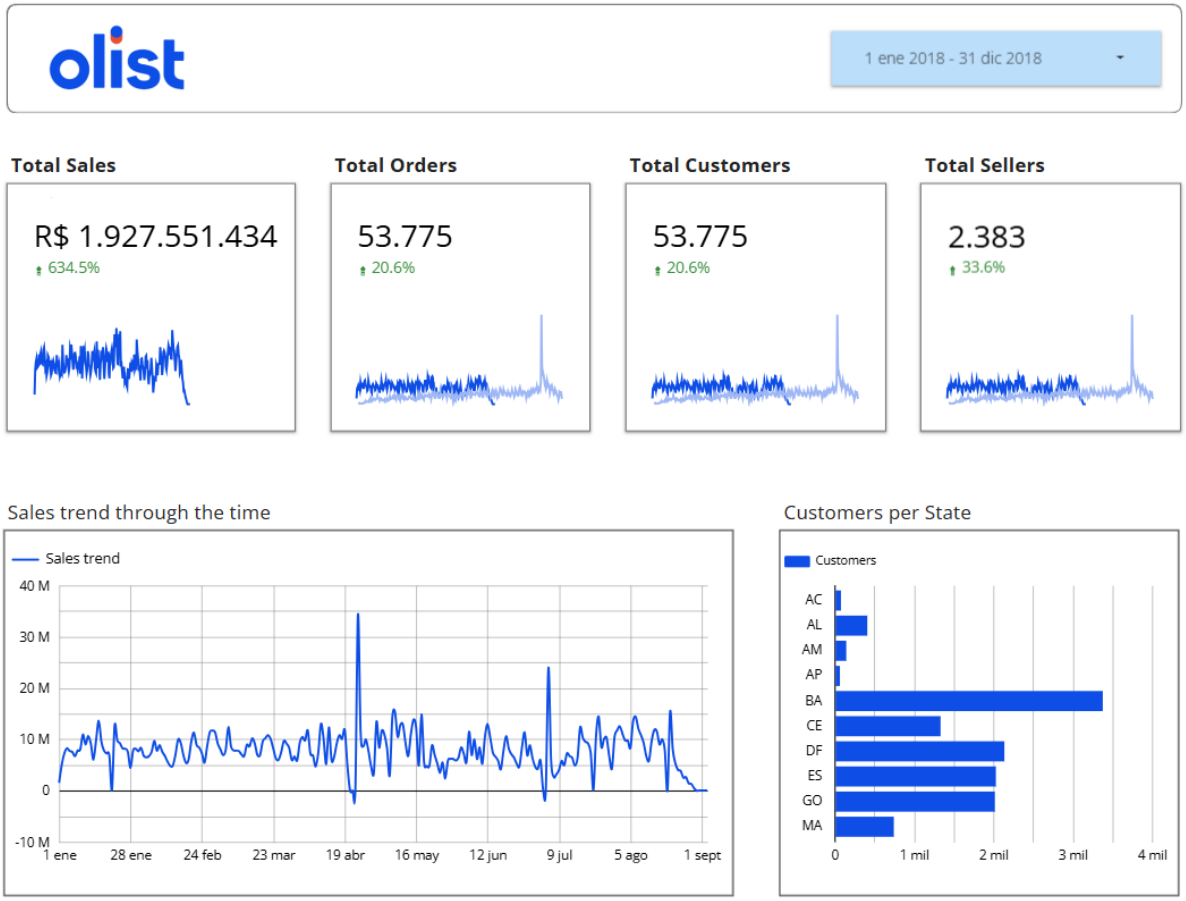
6. Looker Studio Dashboard

The functional dashboard provides real-time visibility into Olist's performance.

https://lookerstudio.google.com/reporting/e29ee641-3674-4448-83a7-16e4dcb2e288



6.1 Key Visualisations

- **Sales Performance Overview:** A scorecard showing total GMV, average order value, and total units sold.
- **Logistics Heatmap:** A map using Dim_Geolocation to visualize regions with the highest delivery delays.
- **Order Status Distribution:** A pie chart showing the percentage of "Delivered" vs. "Pending" orders.
- **Interactive Filters:** A date range picker and a "Product Category" dropdown to allow users to drill down into specific segments.

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7. Data Governance, Metadata & Data Quality

7.1 Governance & Privacy

Olist must comply with **LGPD (Brazil's General Data Protection Law)**[39]. Personal data, such as customer_unique_id, will be protected using hashing during the transformation phase.

7.2 Data Quality and Metadata

- **Master Data Management:** Dim_Products serves as the single source of truth for all product-related reporting across the organisation.
- **Quality Rules:** Automated checks will ensure that price and freight_value are always non-negative and that all order_id records in the fact table have corresponding dimension entries.
- **Lineage:** We will use **Google Cloud Data Catalog** to track metadata and ensure full visibility into the data journey from MySQL to the final dashboard.

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8. Future Predictive Analytics Opportunities

Building upon this foundation, Olist can explore advanced analytics using **BigQuery ML (BQML)**:

1. **Demand Forecasting:** Predicting peak sales periods for specific categories to help sellers optimize inventory levels.
2. **Logistics Prediction:** Developing a model to provide customers with more accurate delivery windows by factoring in historical carrier performance and geographic variables.
3. **Customer Churn Analysis:** Identifying customers likely to stop using the platform based on purchase frequency and recent delivery experiences.

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9. Conclusion

The transformation outlined in this consultancy report marks a pivotal moment for Olist as it evolves from fragmented, operational data management to a unified, analytics-driven enterprise. By migrating to a robust star schema architecture within Google BigQuery, Olist now possesses the infrastructure to analyze logistics, sales, and seller performance at scale, unlocking new levels of speed, accuracy, and flexibility in business intelligence.

This modernization is not merely a technical upgrade—it is a strategic enabler. The adoption of a cloud-native ELT pipeline, modular data modelling, and serverless analytics tools empowers Olist to:

- **Respond rapidly to market changes:** Real-time data ingestion and dashboarding ensure that decision-makers have up-to-date insights for agile responses.
- **Drive operational excellence:** Advanced analytics on logistics, seller performance, and payment flows help identify bottlenecks, optimize delivery, and enhance customer satisfaction.
- **Ensure compliance and governance:** Automated data quality checks, metadata management, and privacy protections (such as LGPD compliance) safeguard sensitive information and support regulatory requirements.
- **Scale effortlessly:** Serverless architecture and modular design allow Olist to grow its data ecosystem without manual intervention or costly infrastructure changes.

Looking ahead, Olist is well-positioned to leverage predictive analytics and machine learning. The foundation laid by this modernization enables the development of models for demand forecasting, logistics prediction, and customer churn analysis—turning data into a strategic asset for innovation and competitive advantage.

In summary, this retail analytics solution is more than a technical blueprint; it is a catalyst for business transformation. By integrating best-in-class cloud technologies, rigorous data governance, and advanced analytics capabilities, Olist empowers its leadership to make data-driven decisions with confidence, driving sustainable growth in the dynamic Brazilian e-commerce market.